

Joaquín Mir Macias

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PROFESSIONAL SUMMARY

Mathematical and AI Engineer currently working as a **Forward Deployed Engineer at Accenture**. Pursuing an **MSc in Advanced Artificial Intelligence**, with hands-on experience in **Python and Machine Learning**, payments analytics and an **internal generative AI pilot at BBVA**.

EDUCATION

- M.Sc. Advanced Artificial Intelligence - GPA: 7.7/10**
Pontifical Comillas University – ICAI

Madrid, Spain
September 2025 – Present

 - Relevant coursework:** Deep Generative Models, Deep Reinforcement Learning, Geometric Artificial Intelligence, Probabilistic Artificial Intelligence, Explainable artificial intelligence (XAI), MLOps and Advanced Natural Language Processing
- Bachelor of Mathematical Engineering and Artificial Intelligence**
Pontifical Comillas University – ICAI

Madrid, Spain
September 2021 – May 2025

 - Relevant coursework:** Machine Learning, Differential Equations, Natural Language Processing, Computer Vision, Discrete Mathematics, Optimization and Simulation, Time Series Forecasting, Deep Learning, Reinforcement Learning and Cybersecurity

RESEARCH & AI PROJECTS

- Detection of Kidney Tumors and Lesions in CT - Bachelor's Thesis - Grade: 9.5/10**
*Collaboration with the **Startup SYCAI Medical**, Barcelona, Spain*

Barcelona, Spain
September 2024 – May 2025

 - Built and evaluated **8 deep learning architectures** (2D/3D U-Net, U-Net 3D Residual, nnU-Net in 2D/3D/cascade, Rel-UNet, and MONAI Auto3DSeg) for automatic kidney-tumor segmentation in CT, on a consolidated real-world dataset of **1,299 scans** (KiTS19/21/23).
 - Designed a reproducible evaluation pipeline (Dice, Surface Dice, HD95, stratified 60/20/20 split) where the best model (nnU-Net cascade) reached a **tumor Dice of 0.85**; also integrated **uncertainty quantification** (Rel-UNet) and **explainability** (Grad-CAM) to flag low-confidence regions for clinical review.
- Project Member - Autonomous Wheelchair - 1st Place, University Competition**
Pontifical Comillas University

Madrid, Spain
October 2023 – June 2024

 - Integrated sensors and ML-based perception models for indoor navigation, preprocessing data within a multidisciplinary engineering team.
- Recent Projects**

 - Algebraic Reasoning Distiller (SAIR Mathematics Challenge) PyTorch, Qwen2.5-7B + LoRA, SFT → GRPO, RAG (Chroma + SBERT). **Multi-agent system** (symbolic prover + RAG retriever + LLM distiller); inference **≤ \$0.01/problem** via ≤10 KB offline cheat sheet.
 - R2-Dreamer - World Models for Visual Continuous Control (PyTorch, RSSM (32×16 categorical), Block-GRU, Actor-Critic, Imagination. Re-implemented R2-Dreamer (**ICLR 2026**) without pixel decoder on CarRacing-v3 / Hopper-v5 / Walker Walk from raw pixels.
 - Uncertainty-Aware Brain Tumor Segmentation - PyTorch, U-Net, MC Dropout, Deep Ensembles, VIMH, SSN, Last-Layer Laplace. Benchmarked 8 uncertainty quantification methods on BraTS 2018 dataset.
 - Digital Twin with LLMs – M.Sc. Final Thesis (in progress)

WORK EXPERIENCE

- Accenture - AI Native Software Engineer**
Forward Deployed Engineer

Madrid, Spain
September 2026 – Present

 - Design and implement LLM-based agents for enterprise clients: **tool use**, retrieval (**RAG**), orchestration and evaluation, from prototype to production.
- BBVA - AI & Data Analytics Intern**
Internal Audit

Madrid, Spain
February 2025 – August 2025

 - Led the internal pilot of Generative-AI tools** (GPT-4-class assistants, Gemini) within Internal Audit: identified concrete audit use cases, drafted a **risk-evaluation framework for AI adoption in banking**, and raised cross-team awareness of AI-specific risks.
 - Automated recurring control tests by building Python data pipelines and scripts, reducing manual workload on standard quarterly reviews.
- Redsys - Data Analyst Intern**
Analytics Systems / Business Intelligence Department

Madrid, Spain
June 2024 – August 2024

 - I Processed **large-scale payments data** using SQL and Hue and delivered an **interactive Qlik Sense dashboard** for acquiring and issuing transactions, defining KPIs with stakeholders and training key users.

ADDITIONAL EDUCATION & LANGUAGES

- International:** 2-week English immersion in Ireland. Hosted an exchange student from the International School of Indiana.
- Online Courses:** Machine Learning (Stanford University); Frontier Transformation Engineer (Microsoft)

TECHNICAL SKILLS

- ✓ Programming languages: **Python** (advanced), **SQL**, JavaScript, MATLAB

✓ ML Frameworks: **PyTorch**, Hugging Face Transformers, LangChain, Scikit-learn, NumPy, Pandas

✓ Languages: **Spanish** (Native), **English** (C1 – TOEFL IBT)

✓ Areas of Expertise: Deep Learning, Computer Vision, NLP, Generative Models (Diffusion, LLMs), Reinforcement Learning, Geometric AI, Probabilistic AI, Explainable AI, AI Agents.

✓ MLOps & infrastructure: **Docker**, Git, MLflow, Kubernetes, Google Cloud

INTERESTS

- Applied AI research, AI Agents, Financial technology and AI for medical health